🏛️ PIXIU TECH / ENGINEERING STRATEGY

Proprietary AI Copilot — Phase 1 Architecture & Component Specification

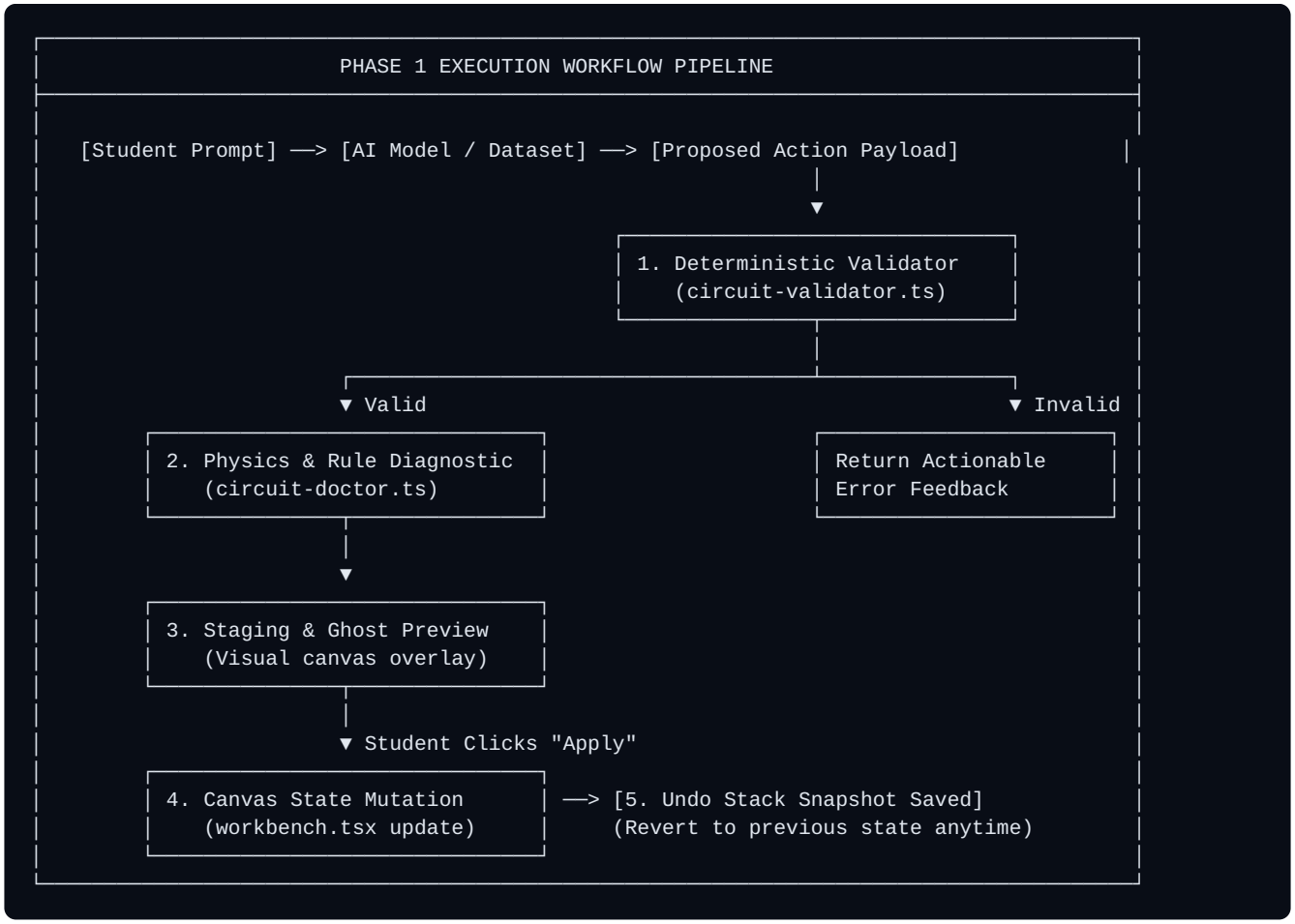
Document Version: 1.0.0 | **Status:** Ready for Review & Enhancement
Classification: Confidential & Proprietary — Pixiu Tech Intellectual Property

📌 Executive Summary

This blueprint defines the engineering execution plan for **Phase 1: AI-Ready CyberLab Foundation**, in direct alignment with Pixiu Tech's *Proprietary AI Copilot Strategy*.

The Core Architectural Principle

> ***"Make the model propose; make software verify."***
> *The AI model is an untrusted agent. It never mutates the live canvas directly. It proposes an action payload; deterministic application software validates the schema, pins, and physics rules; the student previews the changes; and only upon confirmation is the circuit applied with an instant **Undo/Rollback** safety net.*



🗄️ Section 1: Ground-Truth 32-Component Training Specification

Every component in Pixiu CyberLab has been cataloged into a normalized machine-readable dataset: * **TypeScript Definition:** `pixiu-lms-portal/src/lib/ai/pixiu-components-dataset.ts` * **Raw JSON Catalog:** `pixiu-lms-portal/src/lib/ai/pixiu-components-dataset.json`

1.1 Microcontrollers (4 Target Boards)

Board Type	Logic Level	ADC Resolution	Power Output	Key Header Pins & Capabilities	Primary Wiring Rules
`arduino-uno`	5V	10-bit (0–1023)	5V, 3.3V, VIN	D0–D13 (PWM: D3, D5, D6, D9, D10, D11), A0–A5 (I2C: A4=SDA, A5=SCL), 2x GND	All active components must share common GND with Uno. Servos require PWM pins.
`esp32-devkit`	3.3V	12-bit (0–4095)	3.3V, VIN (5V)	GPIO2 (Built-in LED), GPIO4, GPIO15, GPIO21 (SDA), GPIO22 (SCL), GPIO34/35 (Input-only ADC)	Max input 3.3V on GPIO pins. 5V sensors require voltage dividers.
`rpi-pico`	3.3V	12-bit (0–4095)	3.3V, VBUS (5V)	GP0 (SDA), GP1 (SCL), GP25 (Built-in LED), GP26 (ADC0)	Native 3.3V logic level throughout.
`microbit-v2`	3.3V	10-bit (0–1023)	3V, GND	P0, P1, P2 (Touch/ADC/Audio), Btn A, Btn B, 5x5 LED Matrix	Large rings suited for alligator clips. Audio tone defaults to P0.

1.2 Passive & Discrete Components (6 Parts)

Part Type	Default Props	Pins & Recognized Aliases	Electrical Constraints	Mandatory Circuit Doctor Rules
`led`	`color: "red"`	`anode` (`a`, `pos`, `+`), `cathode` (`k`, `neg`, `-`)	1.8V–2.2V drop, max 20mA	**Must** have 220Ω–1kΩ series resistor to prevent burnout. Reverse bias prevents conduction.
`resistor`	`resistance: 220`	`a` (`t1`, `1`), `b` (`t2`, `2`)	Linear Ohm's Law, non-polarized	Symmetrical terminals. Connects in series with LEDs or in voltage dividers.
`pushbutton`	`pressed: false`	`p1a` (`1a`), `p1b` (`1b`), `p2a` (`2a`), `p2b` (`2b`)	Momentary tactile switch	Recommended with `INPUT_PULLUP` to prevent floating states.
`potentiometer`	`value: 512`	`t1` (5V), `wiper` (`w`, `out`), `t2` (GND)	10kΩ linear track	Wiper must connect to an Analog ADC pin (A0–A5).
`battery`	`voltage: 9`	`pos` (`+`), `neg` (`-`)	9V DC external power	Negative terminal must connect to common circuit GND when driving shared loads.
`breadboard`	N/A	`pos_top`, `neg_top`, `pos_bot`, `neg_bot`, columns `a1–j20`	Dual power distribution rails	Distributes 5V and GND buses across multiple components.

1.3 Sensors & Input Devices (10 Parts)

Sensor Type	Output Kind	Pins & Connections	Read Block / C++ API	Application Topology
`ultrasonic`	Digital Pulse	`vcc` (5V), `trig` (D-Pin), `echo` (D-Pin), `gnd` (GND)	`ultrasonic_read`	Sonar distance measurement (2–400 cm).
`ldr`	Analog Passive	`t1` (5V), `t2` (Junction to A0 + 10kΩ to GND)	`ldr_read`	Light-dependent voltage divider. Resistance drops with light.
`pir-sensor`	Digital TTL	`vcc` (5V), `out` (D-Pin), `gnd` (GND)	`pir_motion`	Pyroelectric motion detection. Goes HIGH upon human movement.
`dht11`	Digital Serial	`vcc` (5V/3.3V), `data` (D-Pin), `gnd` (GND)	`dht_temperature`, `dht_humidity`	Climate sensing (0–50°C temp, 20–90% humidity).
`tmp36`	Analog Voltage	`vcc` (5V), `out` (ADC Pin), `gnd` (GND)	`tmp36_temperature`	Linear 10 mV/°C analog output with 500 mV 0°C offset.
`mq2-gas`	Dual (AO/DO)	`vcc` (5V), `gnd` (GND), `do` (D-Pin), `ao` (ADC Pin)	`mq2_gas_read`	Gas & smoke detection. Internal heater requires 5V rail.
`soil-moisture`	Analog Voltage	`vcc` (5V/3.3V), `ao` (ADC Pin), `gnd` (GND)	`soil_moisture_read`	Volumetric soil conductivity probe (high=dry, low=wet).
`ir-sensor`	Digital Active-Low	`vcc` (5V), `out` (D-Pin), `gnd` (GND)	`ir_read`	Short-range reflective optical obstacle detector.
`joystick`	Dual Analog + D	`gnd`, `vcc`, `vrx` (A0), `vry` (A1), `sw` (D2)	`joystick_x`, `joystick_y`, `joystick_button`	Dual 10kΩ potentiometers + momentary tactile push button.
`rotary-encoder`	Quadrature Gray	`gnd`, `vcc`, `clk` (D2), `dt` (D3), `sw` (D4)	`encoder_read`	Continuous rotational dial with push switch.

1.4 Actuators, Motors & Drivers (8 Parts)

Actuator Type	Control Signal	Pins & Wiring	Code Interface	Hardware Safety Rules
`servo`	50Hz PWM Pulse	`sig` (PWM Pin: D9/D10), `vcc` (5V), `gnd` (GND)	`servo_write` (`<Servo.h>`)	Must connect to PWM-capable pin requires 5V rail.
`buzzer`	Audio Frequency	`pos` (D-Pin: D8/D11), `neg` (GND)	`tone_play` , `tone_stop`	Driven by `tone(pin, fre` square wave.
`dc-motor`	High Current DC	`pos` (M+), `neg` (M-)	`motor_driver_write`	**Forbidden** to connect directly to MCU GPIO. Requires L298N o relay.
`relay`	Digital Active-High	`vcc` (5V), `gnd` (GND), `in` (D-Pin), `com` , `no` , `nc`	`io_digitalwrite`	Opto-isolated coil switches external battery to load.
`l298n`	Dual PWM + Dir	`vs` (9V Bat), `gnd` , `5v` , `ena` , `in1` , `in2` , `enb` , `in3` , `in4`	`motor_driver_write`	Common GND between battery a Arduino is strictly mandatory.
`rgb-led`	3x PWM Channels	`red` (D9), `gnd` (Common Cathode), `green` (D10), `blue` (D11)	`io_analogwrite`	Each of the 3 colo anodes requires it own 220Ω resistor
`npn-transistor`	Current Base	`emitter` (GND), `base` (via 1kΩ to D-Pin), `collector` (Load)	`io_digitalwrite`	Base pin must always have 1kΩ resistor to protect MCU GPIO.
`keypad-4x4`	Matrix Scanning	Rows: `r1-r4` (D2–D5), Cols: `c1-c4` (D6–D9)	`keypad_read` , `keypad_check_pin`	Scans 16 keys across 8 digital lin using `<Keypad.h>` .

1.5 Displays (3 Modules)

Display Type	Bus / Interface	Pins	Default Props	Code Libraries
`lcd-i2c`	2-Wire I2C (0x27)	`gnd`, `vcc` (5V), `sda` (A4), `scl` (A5)	`line1: "PIXIU CYBERLAB"`, `line2: "SYSTEM READY"`	`<LiquidCrystal_I2C.h>`, `<Wire.h>`
`oled-i2c`	2-Wire I2C (0x3C)	`gnd`, `vcc` (3.3V/5V), `scl` (A5), `sda` (A4)	`line1: "PIXIU TECH"`, `line2: "READY"`	`<Adafruit_SSD1306.h>`, `<Adafruit_GFX.h>`
`neopixel-ring`	800kHz Single-Wire	`di` (D6), `vcc` (5V), `gnd`, `do`	`numPixels: 8, brightness: 200, pattern: "rainbow"`	`<Adafruit_NeoPixel.h>`
`seven-segment`	Parallel 7 Segments	Segs `a-g` (D2–D8), `dp`, `com` (GND)	`digit: 0`	Direct GPIO bit-mapping

🛠️ Section 2: Phase 1 Deliverables & Technical Architecture

Deliverable 1: Deterministic Circuit Validator (`circuit-validator.ts`)

- **Path:** `pixiu-lms-portal/src/lib/circuit-validator.ts`
- **Contract:** `ts export interface CircuitValidationResult { valid: boolean errors: string[] warnings: string[] normalizedCircuit?: CircuitState metrics: { partCount: number wireCount: number unconnectedPins: number powerRailsOk: boolean } } export function validateProposedCircuit(candidate: unknown): CircuitValidationResult`
- **Validation Rules Executed:**
- **Schema Check:** Verifies `parts` array contains valid coordinates, rotations, and IDs; `wires` connect valid `{partId, pinId}` endpoints.
- **Pin Resolution:** Replaces aliases (e.g., `resistor:1` $\to$ `resistor:a`, `led:+` $\to$ `led:anode`).
- **Short-Circuit Detection:** Blocks direct `5v` $\to$ `gnd` wire connections without load resistance.
- **Burnout Prevention:** Rejects direct `LED` to power pin without an intervening resistor.
- **Ground Loop Continuity:** Flags any active actuator or sensor missing a return ground path.

Deliverable 2: Action & Staging Controller (`use-circuit-copilot-action.ts`)

- **Path:** `pixiu-lms-portal/src/components/circuit-lab/use-circuit-copilot-action.ts`
- **Features:**
- **Staging Mode (`preview`):** Visual ghost preview displaying proposed components and wire trajectories with dotted glow strokes before touching canvas state.
- **Application Mode (`apply`):** Safely merges or replaces parts/wires into `workbench.tsx`'s `updateState`.
- **Code Hydration**: Injects matching Blockly XML or Arduino C++ sketch simultaneously.
- **Undo Stack (`undo`):** Preserves exact previous `(parts, wires, blocklyXml)` snapshot in a circular buffer, enabling one-click reversion.

Deliverable 3: Simulator Telemetry Serializer (`circuit-telemetry.ts`)

- **Path:** `pixiu-lms-portal/src/lib/circuit-telemetry.ts`
- **Features:**
- Extracts live nodal voltages from `solveCircuitNodalPotentials`.
- Reads current MCU pin states (`pinStatesRef.current`), PWM duty cycles, and buzzer frequencies.
- Packages Circuit Doctor health reports into a compact prompt injection context (~120 tokens) so the AI tutor answers physics questions with factual accuracy.

Deliverable 4: Automated Verification Test Suite (`test-phase1-ai-copilot-engine.mjs`)

- **Path:** `scripts/test-phase1-ai-copilot-engine.mjs`
- **Test Gates:**
- **Gate 1:** All 65 existing starter templates pass the validator with 100% compliance.
- **Gate 2:** Intentionally faulty circuits (e.g., invalid pin `uno:d99`, missing LED resistor, floating ground) are caught and rejected with descriptive error strings.
- **Gate 3:** Alias normalization succeeds for all 32 component pin variations.
- **Gate 4:** Preview $\to$ Apply $\to$ Undo state lifecycle maintains state purity without memory leaks.

📊 Section 3: Benchmark & Quality Evaluation Matrix

As established in your engineering strategy guide, progress will be measured by **functional product benchmarks**, not training loss:

Evaluation Metric	Target Threshold	Validation Method
Schema Validity Rate	$\ge 99.5\%$	JSON schema parsing & TypeScript interface compliance.
Circuit Validity Rate	$\ge 98.0\%$	Valid board, part, pin, and alias existence checks.
Simulation Success Rate	$\ge 95.0\%$	Solver runs without nodal divergence or short-circuit crashes.
Code Compilation Rate	$\ge 95.0\%$	Arduino sketch / Blockly AST compiles cleanly.
Diagnostic Repair Rate	$\ge 90.0\%$	Successfully diagnoses and repairs seeded wiring faults.

🚀 Section 4: Phase 1 Action Plan & File Inventory

File Path	Action	Description
`pixiu-lms-portal/src/lib/ai/pixiu-components-dataset.ts`	**Completed**	Full 32-component training & rule specifications.
`pixiu-lms-portal/src/lib/ai/pixiu-components-dataset.json`	**Completed**	Machine-readable JSON export for training pipelines.
`pixiu-lms-portal/src/lib/circuit-validator.ts`	**To Build**	Deterministic schema, pin, and electrical validation engine.
`pixiu-lms-portal/src/lib/circuit-telemetry.ts`	**To Build**	Live simulator telemetry context serializer for AI tutor.
`pixiu-lms-portal/src/components/circuit-lab/use-circuit-copilot-action.ts`	**To Build**	Canvas staging, preview, apply, and undo controller.
`scripts/test-phase1-ai-copilot-engine.mjs`	**To Build**	Automated end-to-end verification gate for Phase 1.

This blueprint establishes the foundation for Pixiu Tech's proprietary AI engine.